Lakshman Rao Botuku

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PROFESSIONAL SUMMARY

AI & Machine Learning Developer with strong foundation in **Python, NLP, backend APIs, and ML model development**. Hands-on experience in building ML models, REST APIs, and deploying data-driven applications using Django and FastAPI concepts. Familiar with **LLMs, RAG pipelines, vector search, OCR, and speech processing**. Passionate about building scalable AI systems and continuously learning advanced AI technologies.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript

AI / Machine Learning & NLP: Scikit-learn, TensorFlow, HuggingFace Transformers, spaCy, IndicBERT (basic), OpenAI GPT (basic usage)

LLM & RAG Frameworks: LangChain, LlamaIndex (exposure), Prompt Engineering

Backend & APIs: FastAPI(basic), Django, Django REST Framework, RESTful APIs, JSON, Postman

Frontend: React.js, Tailwind CSS

Databases & Vector Stores: MySQL, Postgre sql, MongoDB, Firebase

Tools & Platforms: Git, GitHub, Linux, VS Code, PyCharm, Jupyter Notebook, Google colab.

Development Methodologies: MVT Architecture, Responsive Web Design, OOP Principles

Core Concepts: AI Pipelines, Model Inference, Semantic Search, System Design, Data Processing, Scalable Architecture

WORK EXPERIENCE

Quantum Works Private Limited — Software Developer **SEP 2025 – Present**

Key Projects

Enterprise Document Question Answering System (LLM + RAG) - **Dec 2025 – Jan 2026**

- Designed and developed an end-to-end Retrieval-Augmented Generation (RAG) system for answering questions from enterprise documents.
- Implemented document ingestion, chunking, and embedding pipelines using LangChain and HuggingFace embeddings
- Stored vector embeddings in FAISS / Chroma for efficient semantic search and similarity retrieval
- Integrated LLMs (OpenAI GPT / LLaMA / Mistral) to generate accurate, context-aware responses from retrieved documents.
- Built backend inference APIs using FastAPI to serve real-time question answering requests.
- Implemented metadata filtering and relevance ranking to improve answer precision and reduce hallucinations

Labour Management System – Labour Availability & Management System **Oct 2025 – Dec 2025**

- Designed and developed a **Labour Availability & Management System** to manage workers, contractors, and job assignments.
- Built backend APIs using **Django REST Framework** for worker registration, job posting, attendance, and wage tracking.
- Implemented **role-based authentication and authorization** for Admin, Contractor, and Labour users.
- Integrated **PostgreSQL** for transactional data storage and reporting
- Designed scalable system architecture to support **future AI modules** such as worker recommendation and demand forecasting
- Developed clean REST endpoints following **microservice and modular design principles**.

EDUCATION

Narasaropeta Engineering College **2020 – May 2024**
Bachelor of Technology CSE-(Artificial Intelligence) *CGPA 8.37*

Projects

Laptop Price Prediction – Machine Learning

- Designed and implemented an end-to-end **regression pipeline** for predicting laptop prices based on hardware and brand features.
- Performed **data cleaning, outlier handling, encoding, and feature scaling** using Pandas and NumPy.
- Engineered key features including **processor type, RAM, storage, GPU, screen resolution, and brand** to improve prediction accuracy.
- Trained and evaluated multiple models using **Linear Regression, Random Forest, and Gradient Boosting** with cross-validation.
- Achieved **~89% prediction accuracy** with optimized hyperparameters using **GridSearchCV**.
- Analyzed model performance using **RMSE, R^2 score, and residual analysis** to minimize prediction error.
- Deployed the trained model through a **Django web application** with REST endpoints for real-time price prediction.